Base Drag Dominates the Apogee Error Budget in Open-Source Supersonic Rocket Simulation: A Mechanism-Attribution Study with External Base-Pressure Benchmarks

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Abstract

Base drag is the least-constrained term in slender-body supersonic aerodynamics, yet its weight in an *integrated* apogee prediction is rarely isolated. We ask how heavily the base-drag term weighs on the apogee of fin-stabilized supersonic sounding rockets relative to the other supersonic submodels, and whether the closure that carries that weight can be anchored to external base-pressure data. The base-drag closure as implemented—turbulent Chapman–Korst shear-layer theory, the Chapman laminar correlation [1], an ESDU 77021 [2] boundary-layer-thickness correction, the empirical $C_{d,\text{base}} = 0.064 + 0.186/M^2$ correlation, a Viswanath [3] boattail correction, and a finned-base augmentation term—is benchmarked against external, out-of-sample base-pressure measurements, and a controlled mechanism ablation then disables each drag contribution in isolation across 24 of the 25 corpus flights (the MESOS 293K case excluded; 23 single-stage plus the AeroPac 104K two-stage closure) spanning Mach 0.54–3.46 [4]. The turbulent closure reproduces NACA TN 3393 [1] base pressures to a mean absolute percentage error of 15.9% (the Hart NACA RM L52E06 free-flight subset to 4.0%) and the Chapman laminar branch to 4.4%. The ablation shows that base-drag augmentation moves the mean apogee error by 8.10 percentage points—the dominant mechanism, roughly nine times the next supersonic submodel (Van Driest II compressible skin friction, 0.87 pp [5]) and about 54 times the shock-geometry pre-pass (0.15 pp). A decontaminated prospec-

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tive holdout (mean absolute error 3.95% versus 5.47% on development flights) shows that the corpus-frozen base-drag constants generalize rather than overfit. We frame these results as a mechanism-attribution study, backed by external base-pressure benchmarks, that locates base drag at the top of the apogee error budget—not a per-closure apogee intercomparison, and not an integrated-validation or parity claim—complementing the integrated study reported separately [6].

Keywords: base drag, boattail, supersonic rocket, trajectory simulation, mechanism ablation, Chapman–Korst, apogee prediction

1. Introduction

For a slender finned rocket flying supersonically, the pressure deficit acting over the blunt aft face — the base drag — is not a small correction. Once the boundary layer separates at the base and the recirculating wake closes behind a free shear layer, the base pressure falls well below ambient, and the resulting axial force is a large fraction of the total zero-lift drag: of order one quarter to one half across the low-supersonic band where most amateur and sounding-rocket flights spend the bulk of their powered and coasting time. The contribution peaks near $M \approx 1.05$, where the base-drag coefficient of a flat-based body reaches roughly $C_{d,\text{base}} \approx 0.25$, and then decays as the freestream entrains the wake at higher Mach number. Because this term is both large and concentrated in the speed range that dominates the climb-to-apogee integral, the choice of base-drag closure is one of the most consequential modeling decisions in a supersonic trajectory code.

It is also one of the least settled. Base pressure resists first-principles prediction — it depends on boundary-layer state at separation, base geometry, afterbody taper, and, for a finned vehicle, on the fins themselves — so the engineering literature offers a proliferation of competing closures rather than a single accepted law. Chapman's free-shear-layer analysis and the Chapman–Korst recompression model underpin the turbulent base-pressure estimates that remain the reference for nonlifting bodies of revolution [1]; the ESDU 77021 data item consolidates a broad experimental corpus into a recommended base-pressure correlation for supersonic and hypersonic Mach numbers [2]; and practical rocketry codes often fall back on compact empirical fits such as $C_{d,\text{base}} = 0.064 + 0.186/M^2$, which captures the observed high-Mach decay with two coefficients. Layered on top of these are geometry-specific modifiers: a boattail reduces base area and turns the flow, cutting base drag by a substantial margin for well-chosen aft-taper angles [3], whereas a blunt base does not; and a four-fin tail can either relieve or augment the base suction relative to the clean-body value, depending on how the fin wakes and the body wake interact. The closures disagree not only in magnitude but in functional form, and each was calibrated on a particular geometry family. High-fidelity computation has since resolved the near-wake structure in detail — thin-layer Navier–Stokes solutions recover the transonic base flow of an ogive-cylinder-boattail projectile, for instance [7] — but such methods are too costly to embed inside a six-degree-of-freedom trajectory integrator that must evaluate aerodynamic coefficients thousands of times per flight, so engineering codes still rely on the semi-empirical closures above. The open question this paper addresses is therefore not which closure is most physically complete, but *how much* the base-drag term as a whole weighs on the single number a designer cares about — the predicted apogee — relative to the other supersonic submodels, and whether the closure that carries that weight can be anchored to external base-pressure measurements rather than to the flight corpus alone.

Despite the maturity of the individual closures, the open literature does not quantify, in a reproducible way, how heavily the base-drag term weighs in a full six-degree-of-freedom flight against the other supersonic submodels — the single number a designer ultimately cares about being the predicted apogee. Component-level validations report base-pressure or base-drag error against wind-tunnel data, but a base-drag coefficient that matches a tunnel point to within a few percent may still move apogee by far more, or far less, than its component error suggests, because apogee integrates the drag history over a trajectory weighted toward the lower-Mach portion of the flight. Conversely, a submodel with a larger component error can be nearly inert at apogee if its disagreement falls outside the speed range that dominates the integral. The established open-source simulator OpenRocket [8] did not model supersonic base drag at all, while the commercial reference RASAero II [4] embeds its base-drag treatment in closed source, so neither tool exposes the per-mechanism apogee sensitivity in a form that can be audited or reproduced.

The companion study [6] established the integrated context for the present work: an open-source supersonic extension of OpenRocket, validated against a 25-flight external corpus spanning Mach 0.54 to 4.33, that achieves apogee-prediction parity with the commercial RASAero II baseline. A mechanism ablation reported in that work isolated the contribution of each supersonic model to the corpus apogee error and produced a striking ranking. Disabling the finned-base augmentation shifted mean apogee error by 8.10 percentage points (up to 39.5 percentage points on the Mach-2.19 A-601 Kinsel flight) — by far the dominant driver of integrated apogee accuracy. The remaining mechanisms were comparatively minor: compressible (Van Driest II) skin friction accounted for 0.87 percentage points, the DATCOM 4.1.5.1 fin wave-drag term for 0.39 percentage points, and the shock-geometry pre-pass — the celebrated architectural innovation of the companion paper — for only 0.15 percentage points, because that pre-pass is inert below Mach 1 and apogee integrates a trajectory dominated by lower-Mach drag. In other words, the base-drag closure, not the shock geometry, governs the apogee error budget.

This paper takes that ablation result as its point of departure and asks a distinct research question: *how much does the base-drag term weigh in the integrated supersonic-rocket apogee error budget relative to the other supersonic submodels, and can the closure that carries that weight be anchored to external base-pressure measurements?* Where the companion paper [6] reports an integrated parity result and treats base drag as one component among many, this paper diverges into a focused mechanism attribution supported by external component benchmarks. The contribution has three parts. First, we describe the base-drag closure as implemented — the Chapman–Korst turbulent shear-layer model [1], the Chapman laminar correlation [9], the ESDU 77021 boundary-layer-thickness correction [2], the empirical $C_{d,\mathrm{base}} = 0.064 + 0.186/M^2$ correlation (presented as an empirical fit validated against the NACA TN 3393 base-pressure data [1] and consistent with ESDU 77021 [2], with no reliance on any unverifiable correlation), the Viswanath boattail correction [3], and a finned-base augmentation term — and we *benchmark* its branches against external, out-of-sample base-pressure data (NACA TN 3393 turbulent and laminar, and the Hart NACA RM L52E06 free-flight set). Alternative literature closures exist for several of these branches, and we note where they differ in form and calibration geometry; we do *not*, however, substitute them one at a time and report per-closure apogee deltas — that closure-swap experiment is outside the scope of this study. Second, we report a controlled *mechanism ablation* that disables each implemented aerodynamic mechanism in isolation across the corpus and quantifies how much each moves integrated apogee, localizing the base-drag term at the top of the error budget. Third, we confront the in-sample exposure of the corpus-frozen base-drag constants head-on with a decontaminated prospective holdout, which tests whether they generalize. The remainder of the paper specifies the closure and its components, details the benchmark and ablation methodology, reports the component-benchmark, ablation, and holdout results, and discusses what the ranking implies for open supersonic trajectory modeling.

2. The Supersonic Base-Drag Closure and Its Components

This section specifies the supersonic base-drag closure as implemented and its component branches. Each branch contributes to the base-drag coefficient $C_{d,\mathrm{base}}$ referenced to the base area $A_{\mathrm{base}} = \pi R_{\mathrm{base}}^2$, so that its contribution to the vehicle axial-force coefficient is $C_{d,\mathrm{base}}\, A_{\mathrm{base}}/A_{\mathrm{ref}}$. For each branch we state its governing equation, its regime of validity, and the published source against which it was assessed. Three of the six components are geometry-independent functions of Mach number and boundary-layer state — the turbulent Chapman–Korst shear-layer model, the Chapman laminar correlation, and the empirical supersonic correlation — consolidated under the ESDU 77021 data item; the remaining three are geometry modifiers applied multiplicatively to the clean-body value: the ESDU 77021 boundary-layer-thickness correction, the Viswanath boattail relief, and the finned-base augmentation. The branches are stitched together below Mach 1.5 by a single C^1-continuous transonic blend, described last. Table 1 collects the inventory. The engineering literature offers *alternative* closures for several of these branches — competing turbulent base-pressure correlations, different boattail-relief models, and several finned-base treatments — that disagree in functional form and were calibrated on different geometry families; we note these alternatives where relevant, but this study describes and benchmarks the closure as implemented rather than substituting alternatives one at a time and reporting per-closure apogee deltas, which is outside its scope.

2.1. Empirical supersonic correlation

The workhorse turbulent closure is the two-coefficient empirical correlation

$$C_{d,\mathrm{base}} = 0.064 + \frac{0.186}{M^2}, \qquad M > 1.3, \tag{1}$$

applicable to a slender axisymmetric afterbody with a turbulent boundary layer at separation. The two coefficients encode the two physical asymptotes of supersonic base flow: the leading constant 0.064 is the nonzero high-Mach floor toward which the base-pressure coefficient asymptotes as the freestream entrains the wake, and the $0.186/M^2$ term captures the decay of the base-pressure deficit with dynamic pressure as Mach number rises. We present Eq. (1) strictly as an empirical fit: it reproduces the turbulent NACA TN 3393 [1] nonlifting-body base-pressure measurements over $M = 2.73$–4.48 to a mean absolute percentage error of 15.9% (and the Hart free-flight subset to 4.0%), and its functional form is consistent with the recommended supersonic base-pressure correlation consolidated in ESDU 77021 [2]. No further provenance is claimed for the two coefficients beyond this benchmark agreement.

2.2. Chapman laminar base drag

For perfect-finish vehicles whose surface is smooth enough that boundary-layer transition is delayed and the layer remains laminar to the base, the turbulent form of Eq. (1) over-predicts the base suction. The Chapman laminar correlation [9] applies instead:

$$C_{p,b,\mathrm{lam}} = \frac{C_{\mathrm{lam}}}{M^2\sqrt{\mathrm{Re}_L}}, \qquad C_{\mathrm{lam}} = 1870, \tag{2}$$

with Re_L the body-length Reynolds number. The $\mathrm{Re}_L^{-1/2}$ scaling follows from laminar free-shear-layer mixing theory — a laminar layer carries less momentum, mixes less, and recompresses to a lower base pressure than a turbulent layer — and the M^{-2} factor again arises from normalizing the base-pressure deficit by freestream dynamic pressure. The constant $C_{\mathrm{lam}} = 1870$ is a geometric-mean fit to the four condensation-corrected laminar TN 3393 points ($M = 2.73$–4.48, $\mathrm{Re}_L \approx 4$–6×10^6), giving a mean absolute percentage error of 4.4% against those data. Applying the turbulent form of Eq. (1) to the same laminar data yields a mean absolute percentage error of 44%: the boundary-layer state at separation is the dominant discriminator between the two branches, which is precisely why a base-drag benchmark cannot treat "the" base-drag closure as a single object. Equation (2) is blended into the turbulent result over $M = 1.3$–2.5 by the Hermite smoothstep $s(t) = 3t^2 - 2t^3$, weighted by the local laminar fraction so that a fully turbulent body never sees the laminar branch; a clamp at the near-vacuum limit $2/(\gamma M^2)$ prevents unphysical growth at low Reynolds number.

2.3. Chapman–Korst turbulent shear-layer model

The Chapman–Korst dead-air recompression theory [1] provides a turbulent base-pressure estimate from boundary-layer edge conditions, used here as the supersonic-side closure that the empirical correlation blends *into*. The supersonic shear layer separating at the base corner reattaches on the wake axis at a recompression point; the balance between corner expansion and shear-layer recompression fixes the base pressure. For a thin turbulent boundary layer the clean-body coefficient follows the same family as Eq. (1) with an added M^{-4} term, calibrated to the turbulent cylindrical-afterbody tabulation of ESDU 77021 [2]:

$$C_{d,\mathrm{base}}^{\mathrm{thin}} = 0.060 + \frac{0.190}{M^2} + \frac{0.005}{M^4}. \tag{3}$$

This reproduces the ESDU anchor values $C_{d,\mathrm{base}} \approx 0.145$, 0.110, 0.085, and 0.070 at $M = 1.5$, 2.0, 3.0, and 5.0, respectively, which agree closely with Eq. (1) for a thin layer — as expected, since both describe the same recompression physics. Equation (3) is the baseline to which the boundary-layer-thickness correction of Section 2.4 is applied. It is blended into the empirical correlation over $M = 1.2$–1.4 by the same smoothstep $s(t)$, so that pure Eq. (1) governs below $M = 1.2$ and the Chapman–Korst form governs above $M = 1.4$.

2.4. ESDU 77021 boundary-layer-thickness correction

ESDU 77021 [2] consolidates a broad experimental corpus of base pressure on bodies of revolution at supersonic and hypersonic Mach numbers without base injection or combustion, and is the reference data item for the turbulent closures above. Beyond fixing the thin-layer baseline of Eq. (3), it supplies the dependence on boundary-layer thickness: a thicker layer entrains more mass into the recirculating wake, raises base pressure, and so *reduces* base drag. The correction is applied as a multiplicative factor on the thin-layer coefficient,

$$f_\delta = 1 - k(M)\sqrt{\theta/R_{\mathrm{base}}}, \qquad k(M) = 0.8 + \frac{0.2}{M}, \tag{4}$$

where θ is the turbulent momentum thickness at the base, estimated from the flat-plate correlation $\theta/x = 0.036\,\mathrm{Re}_x^{-1/5}$, and θ/R_{base} is clamped to $[10^{-5}, 0.5]$ with f_δ clamped to $[0.3, 1.0]$. For a typical rocket ($\theta/R_{\mathrm{base}} \approx 0.01$) the reduction is a few percent; for an unusually thick layer ($\theta/R_{\mathrm{base}} \approx 0.1$) it reaches 20–30%. This correction is the mechanism by which the slender, high-fineness airframes in the corpus carry a distinct base-drag signature from short, blunt bodies, and it is the literature-anchored counterpart to the corpus-frozen thick-boundary-layer amplification disclosed in the companion study [6].

2.5. *Viswanath boattail relief*

When the afterbody tapers inward to a boattail, the converging flow turns through a compression that raises base pressure and shrinks the base area, reducing base drag below the blunt-base value. Following Viswanath [3], the relief is applied as a multiplicative retention factor η_{bt} on the clean-body base drag, parameterized by the boattail half-angle $\theta_{bt} = \arctan\big[(R_{\text{fore}} - R_{\text{aft}})/L_{bt}\big]$ and, in the effective regime, by Mach number:

$$\eta_{bt} = \begin{cases} 0.25 + 0.05\,\theta_{bt}, & \theta_{bt} < 6^\circ, \\ \big[0.55 + 0.04(\theta_{bt} - 6^\circ)\big]\,\big[1 + 0.1\max(0, M-1)\big], & 6^\circ \le \theta_{bt} < 16^\circ, \\ 0.95 - 0.05(\theta_{bt} - 16^\circ), & \theta_{bt} \ge 16^\circ, \end{cases} \tag{5}$$

with η_{bt} clamped to $[0, 1]$. For well-chosen boattail angles of 6–16° this yields a 15–40% base-drag reduction, consistent with the afterbody-drag-management survey of Viswanath [3]. The factor is applied to the wake *downstream* of a boattail and is not also applied on the boattail component itself, which would double-count the relief and can drive the aft base drag to zero on steep imported geometries that still expose a finite blunt exit area.

2.6. *Finned-base augmentation (externally calibrated, B-level)*

The dominant apogee mechanism (§4.2) is the augmentation of base drag by the tail fins. Fins mounted at or just ahead of the base disrupt the recompression of the body wake, deepening the base suction relative to the clean-body value; aeroballistic-range data on the Basic Finner [10] and the afterbody-drag compendium of Hoerner [11] consistently show 40–60% higher base drag for four-fin configurations than for the corresponding smooth cylindrical afterbody over $M = 1.5$–3, which we use as an external *bound* on the plausible magnitude of the effect rather than as a calibration target. The augmentation is applied as a multiplicative factor on the post-relief base drag, written here in its full implemented form,

$$\Phi_{\text{fin}} = 1 + K_{\text{fin}}\,\frac{1 - e^{-n_{\text{fins}}/1.4}}{1 - e^{-4/1.4}}\ \text{clamp}\Big(\frac{s_{\max}}{R_{\text{base}}},\, 0.3,\, 1\Big)\, f(M), \qquad K_{\text{fin}} = 0.55, \tag{6}$$

where the fin count n_{fins} enters through a saturating term normalized to unity at the four-fin reference (wake disruption is not linear in fin count: three corner-wake sectors already nearly divide the base shear layer, so a three-fin body is not under-counted), $s_{\max}$ is the maximum fin span, and the Mach ramp $f(M)$ rises from 0 at $M = 0.2$ to 0.30 at $M = 0.8$, then from 0.30 to 1 over $M = 0.8$–1.3, holds through the low-supersonic band to $M = 3.0$, then decays as $3/M$ above $M = 3.0$ as the fins become progressively submerged in the body shock cone. At the four-fin reference (four rectangular fins, $s_{\max}/R_{\text{base}} = 2.0$, where both geometry factors saturate to unity) the scale constant $K_{\text{fin}} = 0.55$ gives a $\sim 50\%$ augmentation at $M = 2$, inside the 40–60% literature bound above; companion scale constants of 1.00 for rounded-trailing-edge fins and 1.35 for fin-can sleeve geometries cover the remaining corpus afterbody types.

We classify K_{fin} as a *B-level* constant in the specific sense that it lacks validation against an isolated finned-base base-*pressure* dataset. It is *not* corpus-frozen: the value 0.55 is calibrated externally against the Basic Finner aeroballistic-range data [10] so that it reproduces the $\sim 50\%$ augmentation measured there at $M = 2$, which also places it inside the 40–60% over-clean-body band of the same fixture and the Hoerner compendium [11]. Its anchor is thus a published external fixture, not flights drawn from the calibration corpus, so it does not make the headline in-sample. The constant nonetheless does not yet carry the A-level external validation of Eqs. (1) and (2), which are checked against isolated base-pressure data, and this B-to-A gap is the explicit limitation of the augmentation closure. The distinction is material precisely because the mechanism ablation of §4.2 ranks this term as the dominant driver of integrated apogee error, shifting mean corpus apogee error by 8.10 percentage points when removed — about 54 times the 0.15-point effect of the shock-geometry pre-pass. The two genuinely corpus-frozen constants in the base-drag treatment belong to *different* closures — the thick-boundary-layer base-drag amplifier (THICK_BL_K) and the slender-body pressure-drag term (SLENDER_BODY_K) — and it is those two constants, not K_{fin}, that the decontaminated prospective holdout of §4.3 quarantines and shows to generalize (blind holdout MAE 3.95% below development 5.47%). The defense of K_{fin} is instead its external Basic Finner anchor and the component base-pressure benchmarks. Elevating K_{fin} from this total-drag anchor to A-level validation against an isolated finned-base base-pressure dataset is identified in the Conclusions as the single highest-value evidence gap raised by this work.

2.7. *Transonic blend and C^1 continuity*

Below $M = 1.5$ the supersonic closures above are not valid, and the base drag instead rises to a transonic peak near $M \approx 1.05$, where a flat-based body reaches $C_{d,\text{base}} \approx 0.25$. The transonic band $M = 0.85$–1.5 is represented by a degree-4 polynomial anchored at three points: the peak value $C_{d,\text{base}} = 0.25$ at $M = 1.05$, the free-flight plateau $C_{d,\text{base}} = 0.230$ at $M = 1.30$ (just below the Hart NACA RM L52E06 [12] finless free-flight reading of 0.250 ± 0.013, within its digitization uncertainty), and a C^1 handoff to the supersonic correlation at $M = 1.5$. The polynomial is constructed to be continuous in both value and first derivative at the regime boundaries, and the laminar, Chapman–Korst, and boattail blends of the preceding subsections all use the same Hermite smoothstep $s(t) = 3t^2 - 2t^3$, which has zero derivative at its endpoints. This global C^1 continuity is not cosmetic: a discontinuity in $C_{d,\text{base}}$ or its slope near Mach 1 drives the fourth-order Runge–Kutta trajectory integrator to oscillate whenever the vehicle repeatedly crosses the sonic line during boost and coast, and the resulting integration noise would contaminate the very apogee sensitivity this paper sets out to measure. The same continuity discipline is applied across every Mach-regime transition in the underlying simulator, as detailed in the companion paper [6].

3. Benchmark and Mechanism-Ablation Methodology

The methodology has three parts, ordered from the most external evidence to the most in-sample. The *component intercomparison* (Section 3.1) is the substantive physics of the paper: it asks how each candidate base-drag closure reproduces *measured* base pressure in isolation, against external wind-tunnel and free-flight data that played no part in any corpus calibration. The *mechanism ablation* (Section 3.2) is the motivating experiment: it asks how much each implemented aerodynamic mechanism actually moves integrated apogee, by disabling it one at a time across the flight corpus, and it is what singles out base drag as the dominant driver. The *decontaminated holdout* (Section 3.3) is the in-sample defense: it asks whether the two corpus-frozen base-drag scale constants generalize to flights that never informed their calibration, pre-empting the circularity critique that the ablation substrate would otherwise invite. We lead with the externally validated closure fidelity, use the ablation as the finding that motivates a base-drag study at all, and confront the in-sample constants head-on rather than letting them read as the result.

3.1. Component intercomparison protocol

Each base-drag closure introduced in Section 2 is exercised against published base-pressure data with the boundary-layer state and length Reynolds number Re_L matched to the reference condition, so that closures are compared on the flow regime each was derived for rather than averaged across regimes. The discriminator is deliberate: as Section 2.2 shows, applying a turbulent correlation to laminar separation data inflates the error by an order of magnitude, so the protocol conditions closure selection on the boundary-layer state and not on Mach number alone. The error metric throughout is the mean absolute percentage error (MAPE) of the predicted base-pressure or base-drag coefficient against the digitized reference points. Three external datasets anchor the comparison, none of which was used to tune any integrated-trajectory constant; a fourth, computational, comparator is identified but deferred.

NACA TN 3393 (turbulent and laminar).. The primary anchor is the Reller and Hamaker base-pressure investigation of nonlifting bodies of revolution at Mach 2.73–4.98 [1], which reports base pressure for both turbulent and laminar boundary-layer states on the same geometries and therefore exercises both base-drag branches against a single, internally consistent experiment. Against the turbulent points the empirical correlation $C_{d,\mathrm{base}} = 0.064 + 0.186/M^2$ of Eq. (1) attains a MAPE of 15.9%; against the laminar points the Chapman laminar closure of Eq. (2) attains 4.4% [1]. The laminar closure is thus markedly more faithful on its target regime, and the two branches are not interchangeable: applying the turbulent correlation to the same laminar data yields a MAPE of 44%. This is the quantitative basis for treating "the" base-drag closure as a state-dependent family rather than a single object.

Hart NACA RM L52E06 (transonic turbulent).. To extend the turbulent comparison below the TN 3393 floor and through the transonic base-drag peak near $M \approx 1.05$, the turbulent closure is additionally compared against the Hart free-flight base-pressure measurements (NACA RM L52E06 [12]), where it attains a MAPE of 4.0%. The two turbulent anchors together span the transonic peak (Hart) and the supersonic decay (TN 3393), and bracket the band in which the empirical correlation $C_{d,\mathrm{base}} = 0.064 + 0.186/M^2$ and the degree-4 transonic blend of Section 2.7 operate. The Hart finless free-flight reading of 0.250 ± 0.013 near the peak also fixes the transonic anchor point of that blend.

Chapman laminar data.. The laminar branch is anchored on the laminar subset of the same TN 3393 experiment [1], against which the Chapman laminar closure attains the 4.4% MAPE quoted above. This isolates the "perfect-finish" / delayed-transition case relevant to highly polished slender vehicles, for which the turbulent correlation over-predicts base drag, and provides the externally validated counterpart to the laminar branch invoked for the smoothest airframes in the corpus.

Transonic CFD comparator (deferred, disclosed).. A fourth comparison is identified but not executed in this work: the Sahu, Nietubicz and Steger thin-layer Navier–Stokes computation of base flow on a secant-ogive-cylinder-boattail projectile at transonic speed ($M \approx 0.9$–1.2) [7] would provide an independent, CFD-grade check of the closures through the transonic peak, where the empirical and Chapman–Korst forms hand off to the degree-4 blend. Its geometry is structurally different from the projectile families used for the other anchors, so reproducing it would require building a separate vehicle model and digitizing the published base-pressure distributions. We disclose this comparison as *deferred*—it is documented as a memo only and is not exercised as a quantitative comparator here—rather than report a partial result. We claim *no computational fluid dynamics of our own*; [7] is a published third-party computation cited as a prospective comparator.

3.2. Controlled mechanism-ablation protocol

The component intercomparison establishes closure fidelity in isolation but does not establish which mechanism governs the *integrated* apogee error. A closure that matches a wind-tunnel point to a few percent may still move apogee by much more, or much less, than its component error suggests, because apogee integrates the drag history over a trajectory weighted toward the lower-Mach portion of the flight. To resolve this, we run a controlled ablation on 24 of the 25 corpus flights—the externally selected supersonic altitude-comparison set, spanning Mach 0.54–3.46, used for the integrated validation reported in the companion flagship study [6]. The ablation set is the full corpus with the MESOS 293K case excluded, leaving 23 single-stage flights plus the AeroPac 104K two-stage closure. Here the corpus is the *substrate for a mechanism-attribution experiment*, not the basis for an accuracy headline; MESOS 293K is the single case excluded because its high-Mach staging physics confound a single-mechanism attribution.

The protocol disables one aerodynamic mechanism at a time, re-flies all 24 of these trajectories, and records the change in the corpus mean absolute apogee error, $\overline{|\Delta|}$, relative to the fully enabled baseline. Five mechanisms are ablated: the finned-base drag augmentation Φ_{fin} of Eq. (6); the Van Driest II compressible skin-friction transformation [5]; the DATCOM Section 4.1.5.1 fin wave-drag closure [13, 14]; the shock-geometry pre-pass; and the Pitts–Nielsen–Kaattari fin–body interference correction, together with its K_1 floor.

Each mechanism is ablated *in isolation*—disabled singly, with every other mechanism held at its baseline state, and the corpus re-run from a clean configuration for each ablation. This isolation discipline is essential and not incidental: the ablation flags are evaluated as shared static state, so toggling several mechanisms within a single process risks cross-contaminating the recorded baseline and corrupting the attribution. Running each ablation as an independent, clean corpus pass removes that contamination, and the resulting single-mechanism apogee sensitivities are the per-mechanism $\overline{|\Delta|}$ values reported in Section 4.

The ranking that emerges—and the finding that motivates this paper—is steeply ordered. The finned-base augmentation dominates the apogee error budget at a mean $\overline{|\Delta|}$ of 8.10 pp (reaching 39.5 pp on the Mach-2.19 A-601 Kinsel flight), more than an order of magnitude larger than the next mechanism. Compressible skin friction contributes 0.87 pp, the DATCOM fin wave-drag term 0.39 pp, and the shock-geometry pre-pass—the celebrated architectural innovation of the companion paper—only 0.15 pp, because that pre-pass is inert below Mach 1 and apogee integrates a trajectory dominated by lower-Mach drag; the PNK interference term, with its K_1 floor, registers no measurable apogee effect (0.00 pp). Base drag, not shock geometry, governs the apogee error budget, which is what licenses a focused base-drag closure study distinct from the integrated parity result of [6].

3.3. Decontaminated prospective holdout for the corpus-frozen constants

The turbulent base-drag branch carries two scale constants that were frozen against the corpus rather than against an independent component dataset: a thick-boundary-layer scale, `THICK_BL_K` = 2.2, and a slender-body scale, `SLENDER_BODY_K` = 0.0025. Because these constants were tuned on corpus flights, the integrated result is *partly in-sample*, and a naive corpus-wide accuracy figure would be optimistically biased. We disclose this plainly and test generalization with a decontaminated prospective holdout, which is the defense against the circularity critique that follows from using a corpus-calibrated quantity as the ablation substrate of Section 3.2.

The decontamination rule is strict: *every* flight that either constant touched during calibration—the anchor flights for both constants and the source-diagnostic flight—is quarantined in the development partition. This decontamination is applied as an overlay on the corpus's existing development/holdout bookkeeping: the five flights involved in fixing the two constants (Raven, Rabia, Rabia Short Fin Can, Kinsel, and Torrent) are placed in the development partition (DEV) together with the cases already reserved for closure development and ablation, while no calibration-touched flight is allowed to remain in the blind set (HOLDOUT). This yields a development partition of thirteen flights and a holdout of twelve. We stress that this calibration decontamination is a stricter criterion than, and is not to be conflated with, the corpus's prospective forward-freeze protocol, which partitions flights by whether they may be re-tuned after a freeze date rather than by whether they informed the two scale constants, and therefore assigns three of the calibration-touched flights differently. The constants are frozen on DEV and never re-touched; HOLDOUT is scored exactly once. If the constants merely overfit the corpus, HOLDOUT accuracy should degrade relative to DEV; if they capture a transferable physical scaling, HOLDOUT accuracy should hold or improve.

The split yields a development partition of $n = 13$ (mean signed error $+0.22\%$, MAE 5.47%) and a blind holdout of $n = 12$ (mean signed error -1.03%, MAE 3.95%). These partition statistics are computed from the per-flight apogee errors of the archived corpus-baseline artifact, evaluated against the current-code MESOS 293K value, so the split is regenerable from the released data under the membership stated above. The holdout is *more* accurate than the development partition ($3.95\% < 5.47\%$), which is the primary in-sample defense: the two corpus-frozen base-drag constants generalize to unseen flights rather than overfitting the calibration set. This result is what permits the corpus to serve as the ablation substrate in Section 3.2 without circularity, and it is consistent with the holdout reported in the companion study [6].

4. Results

We present the results in three layers that move from the isolated closure to the integrated trajectory. First we benchmark each base-drag branch against the published base-pressure measurement that anchors it (§4.1) — the externally validated, out-of-sample physics on which the comparison rests. Second, we report the controlled mechanism ablation that ranks the integrated-apogee leverage of every drag contribution (§4.2) — the motivating finding that localizes base drag at the top of the error budget. Third, because two of the base-drag closures carry corpus-frozen scale constants, we test directly whether those constants generalize off the development set (§4.3), pre-empting the circularity that an in-sample calibration would otherwise invite.

4.1. Component intercomparison against base-pressure data

Table 2 compares each base-drag closure against the base-pressure data that anchors it; these are external, third-party measurements, not flights from the calibration corpus, and they carry the substantive physics of the comparison. The turbulent Chapman–Korst shear-layer closure reproduces the NACA TN 3393 nonlifting-body base pressures [1] to a mean absolute percentage error (MAPE) of 15.9% over the digitized Mach 2.73–4.48 series, and to 4.0% against the Hart RM L52E06 transonic base-pressure set used to anchor the lower-Mach blend. The Chapman laminar branch [1], applicable to delayed-transition "perfect-finish" bodies, reproduces the TN 3393 laminar series to 4.4% MAPE — nearly four times tighter than the turbulent closure, and a clean discriminator between the two boundary-layer states: applying the turbulent correlation to the laminar data instead degrades the MAPE to roughly 44%. The compact empirical correlation $C_{d,\text{base}} = 0.064 + 0.186/M^2$ is presented not as an independent anchor but as an engineering fit that we verify to be consistent with the TN 3393 turbulent envelope [1] and with ESDU 77021 [2] across the supersonic span; the Viswanath boattail correction [3] is exercised only inside its 6–16° trailing-edge-angle validity window. The 15.9% turbulent residual is the largest of the base-drag errors, consistent with the well-known difficulty of predicting turbulent base pressure — which is sensitive to approach boundary-layer thickness, surface finish, and base geometry detail not fixed by the reference geometry alone — yet the closure reproduces the correct Mach trend and order of magnitude throughout. Published transonic base-flow CFD on a comparable ogive–cylinder–boattail afterbody [7] corroborates the same trend qualitatively.

4.2. Mechanism ablation: base drag dominates the apogee budget

The component MAPEs above quantify each closure's pointwise fidelity but say nothing about its leverage on *integrated* apogee. To isolate that leverage we disabled each drag mechanism in turn and re-ran 24 of the 25 corpus flights (MESOS 293K excluded; 23 single-stage plus the AeroPac 104K two-stage closure), spanning Mach 0.54–3.46, recording the resulting shift in mean absolute apogee error. The mechanisms are mutually orthogonal in the code, so each row of Table 3 is a clean one-at-a-time effect rather than an interaction term.

The ranking is unambiguous. Disabling the finned-base/base-drag augmentation moves the corpus mean apogee error by 8.10 pp — approximately a factor of nine above the next-largest term (Van Driest II compressible skin friction, 0.87 pp [5]), a factor of twenty above DATCOM 4.1.5.1 fin wave drag (0.39 pp [13]), and a factor of ~ 54 above the shock-geometry pre-pass (0.15 pp); the Pitts–Nielsen–Kaattari interference and K_1 floor register no measurable apogee effect at all. Its peak effect, 39.5 pp on the Kinsel $M\,2.19$ flight, exceeds the entire apogee error budget of every other mechanism combined. The shock-geometry pre-pass, by contrast, is inert below $M = 1$ and contributes only 0.15 pp to integrated apogee: apogee integrates a trajectory dominated by lower-Mach drag, so the pre-pass's value lies in local-flow fidelity (correct post-shock conditions for fin loads and stability) and as an architectural seam for downstream supersonic models, not in gross-apogee accuracy. This is the central finding of the paper — among the mechanisms governing supersonic-rocket apogee, the base-drag closure sits at the top of the error budget, while the architecturally prominent shock-geometry pre-pass barely moves the integrated result.

4.3. Generalization of the base-drag constants

Two of the base-drag closures carry corpus-frozen scale constants: THICK_BL_K = 2.2 in the thick-boundary-layer base-drag amplifier and SLENDER_BODY_K = 0.0025 in the slender-body pressure-drag closure. These constants belong to those two secondary closures, not to the finned-base augmentation, whose scale factor $K_{\text{fin}} = 0.55$ is fixed externally against the Basic Finner [10] (§2.6). Because the two corpus-frozen constants are calibrated with reference to flights that also appear in the corpus, the aggregate is in part in-sample, and we test their generalization head-on with a decontaminated prospective holdout: every flight that either constant touched during calibration (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is placed in the development split, leaving a genuinely blind holdout. The development split ($n = 13$) attains a mean absolute apogee error of 5.47% (mean signed +0.22%); the blind holdout ($n=12$) attains 3.95% (mean signed −1.03%). The holdout is *more* accurate than the development set, so the two corpus-frozen constants generalize rather than overfit, and the in-sample calibration does not inflate the headline result. This holdout quarantines those two secondary constants; the dominant finned-base term is defended separately by its external Basic Finner anchor and the component base-pressure benchmarks above.

Taken together, the three layers tell a single story: the base-drag closure is the pointwise-hardest term to predict (15.9% turbulent MAPE against TN 3393 [1]), the highest-leverage term for integrated apogee (8.10 pp ablation effect, $\sim 9\times$ the next mechanism), and — in its two corpus-frozen secondary constants — a generalizing one (blind holdout 3.95% below development 5.47%). We defer the integrated head-to-head against the commercial baseline, and the trajectory-level accuracy statistics, to the companion study [6]; the contribution here is the mechanistic localization of base drag at the top of the supersonic-apogee error budget, together with the external base-pressure validation of the closures that occupy it.

5. Discussion

5.1. Base-drag fidelity is the highest-leverage modeling investment

The central result of the ablation (Section 4) is a sharp ordering of mechanisms by their leverage on integrated apogee. Disabling the finned-base augmentation closure moves the corpus mean apogee error by 8.10 pp, an order of magnitude more than any other supersonic mechanism: Van Driest II compressible skin friction contributes 0.87 pp [5], the DATCOM 4.1.5.1 fin wave-drag term 0.39 pp [13, 14], and the shock-geometry pre-pass only 0.15 pp. The Pitts–Nielsen–Kaattari fin-body interference floor moves apogee by 0.00 pp. The base-flow closure is therefore not merely one term among several: on this corpus it is the single dominant driver of where the integrated trajectory ends up.

This ordering carries a direct prescription for where modeling effort should go. A practitioner extending an open-source trajectory code to supersonic speeds faces a long menu of candidate refinements — higher-order wave-drag theory, boundary-layer transition modeling, crossflow corrections, refined pitch damping — and finite time to implement them. The ablation says plainly that, for the quantity most users ultimately care about, base-flow fidelity is the highest-leverage investment, returning roughly nine times the apogee accuracy per unit of modeling effort as the next-ranked mechanism and about 54 times that of the shock-geometry pre-pass. The closure branches described here — the Chapman–Korst turbulent shear-layer theory [1, 15], the Chapman laminar branch [9], the ESDU 77021 bodies-of-revolution correlation [2], the empirical $C_{d,\text{base}} = 0.064 + 0.186/M^2$ form, and the Viswanath boattail-relief correction [3] — together make up the term that carries this leverage. We do not isolate the apogee contribution of each branch separately; the ablation attributes the leverage to the base-drag term as a whole, and within it the finned-base augmentation is the component the corpus is most sensitive to. The prescription is that refining base-flow fidelity, and especially the treatment of the finned base, returns more apogee accuracy than refining any other supersonic sub-model first.

5.2. Mechanism attribution versus local-flow fidelity

It is tempting to equate architectural prominence with apogee leverage, and the shock-geometry pre-pass is the case where that equation breaks. The pre-pass is the most structurally celebrated component of the supersonic extension: it solves the oblique-shock and Taylor–Maccoll cone field once per timestep and threads the post-shock local conditions into every downstream model. Yet it moves integrated apogee by only 0.15 pp, with a maximum single-flight effect of 3.6 pp (FMJ Black Rock 6, $M = 2.46$). The reason is not that the pre-pass is wrong but that apogee integrates a trajectory whose drag impulse is dominated by the lower-Mach portion of the ascent, and the pre-pass is inert below $M = 1$. A mechanism can be both physically essential and apogee-irrelevant.

The pre-pass earns its place for a different reason: local-flow fidelity. The post-shock conditions it produces feed fin loads and static stability, where they have been verified to match Taylor–Maccoll cone tables [16, 17] to better than 1% and the analytic shoulder-expansion result to numerical precision. It is also the architectural seam that makes the downstream supersonic closures — including the base-flow models studied here, which depend on the local edge conditions at separation — well-posed in the first place. The honest framing is thus twofold: the pre-pass is indispensable for correct *local* aerodynamics (fin loads, static and dynamic stability, recovery-event timing) and as an enabling architecture, but it is not a gross-apogee mechanism.

Conflating the two would misdirect a reader toward refining shock geometry when the apogee error budget sits almost entirely in the base-flow term.

This is precisely why mechanism attribution, rather than aggregate accuracy alone, is the useful product of an ablation study. An integrated parity statistic tells a user that the tool works; it does not tell a developer *which* sub-model to improve next. The decomposition here answers that second question, and its answer — base flow first, everything else second — redirects future modeling and validation effort toward the term that actually controls the trajectory-integrated result.

5.3. In-sample exposure of the finned-base closure, stated plainly

The dominant mechanism is also the one carrying empirical calibration, and intellectual honesty requires confronting that directly rather than burying it. There are two distinct in-sample exposures here, applied by different parts of the code, and they must be separated to be answered honestly.

The first exposure is in the base-flow *scale* constants. Two corpus-frozen boundary-layer scale factors enter the base-drag treatment — THICK_BL_K = 2.2, anchored on the Raven flight, and SLENDER_BODY_K = 0.0025, anchored on Raven, Rabia, and Kinsel. Because those two constants were fitted with reference to flights that also appear in the validation corpus, the headline aggregate is *partly in-sample.* This is the exposure the decontaminated holdout was designed to neutralize, and it neutralizes it directly. Every flight that either scale constant touched — Raven, Rabia, Rabia Short Fin Can, Kinsel, and the Torrent diagnostic case — was placed in the development partition, leaving a genuinely blind holdout. The development partition ($n = 13$) returns a mean signed error of $+0.22\%$ and a mean absolute error of 5.47%; the holdout ($n = 12$), which neither scale constant ever saw, returns mean -1.03% and MAE 3.95%. The holdout is *more* accurate than the development set. Overfit closures degrade on unseen data; a closure whose scale factors improve on the holdout is generalizing the underlying base-flow physics rather than memorizing the calibration flights. This inversion — holdout better than dev — is strong evidence that the two scale constants generalize.

The second exposure is the finned-base augmentation factor itself (FINNED_BASE_K), the constant whose disabling produces the 8.10 pp dominant-mechanism result. We state plainly that the decontaminated holdout above does *not* isolate this factor: the holdout split was constructed to quarantine the two boundary-layer scale constants, not the flights that informed the finned-base augmentation, so it cannot be read as a clean-room test of the dominant lever.

The defense of FINNED_BASE_K therefore rests not on that holdout but on external bounding. The component benchmarks of Section 4 fix the relevant physics from outside the corpus entirely: the turbulent clean-body closures reproduce the NACA TN 3393 nonlifting-body base pressures to 15.9% MAPE (and the Hart free-flight subset to 4.0%) [1], the Chapman laminar branch to 4.4% [1, 9], and the magnitude of the finned-base augmentation itself sits inside the 40–60% over-clean-body band reported in aeroballistic-range testing of the Basic Finner [10] and in Hoerner's afterbody-drag compendium [11]. The augmentation factor is thus fixed by external finned-afterbody data, not by the flight corpus; it does not invent the mechanism. We make no claim that any of these constants are physically derived — they are disclosed calibrations — but the external Basic-Finner anchor of the augmentation factor, together with the holdout generalization of the two corpus-frozen scale constants, is what licenses naming the finned base the dominant apogee driver. Elevating FINNED_BASE_K from its present total-drag anchor to an isolated finned-base base-pressure benchmark is a clean direction for future work.

5.4. Relation to the integrated companion study

The present paper and its integrated-validation companion answer different questions, and the divergence is the point. The question here is one of mechanism leverage — *which* closure governs the trajectory-integrated result — and its answer is that the apogee accuracy rests disproportionately on the base-flow closure while the celebrated shock-geometry architecture is, by the apogee metric, nearly inert. The companion paper [6] asks instead whether the integrated tool predicts supersonic apogee and how it compares to the commercial baseline [4], and reports that integrated-parity result; it does not decompose *why* the prediction lands where it does. The two studies share a corpus and a code base but pursue distinct ends, and the value of reading them together is that the present decomposition explains the structure underlying the companion's aggregate. The claim defended here is a statement about mechanism leverage, not about aggregate parity, and it stands on the ablation and the holdout alone.

5.5. Generality across vehicle classes

How far does "base drag governs apogee" carry beyond the present corpus? The mechanism that makes base drag dominant — a large, separated aft-face pressure deficit acting through the low-supersonic band where the climb-to-apogee integral is weighted most heavily — is generic to slender, fin-stabilized, flat-or-near-flat-base vehicles, the class that spans the corpus from minimum-diameter amateur high-power rockets through university-competition airframes to the single- and two-stage sounding rockets that reach the upper Mach range. Within that class the conclusion is expected to be robust, because every member shares the same qualitative base-flow topology and the same trajectory weighting toward the lower-supersonic regime. The corpus itself spans Mach 0.54 to 4.33 and a wide spread of fineness ratios and fin geometries, and the dominance of the drag-coefficient term is not an artifact of one airframe: a per-flight sensitivity sweep (Fig. 1) shows that the overall drag-coefficient scale — of which base drag is the dominant supersonic component — is the single most leveraged parameter for predicted apogee across structurally diverse vehicles spanning the low-supersonic to upper-Mach range, dwarfing launch geometry and integration-timestep sensitivities on every flight shown.

The conclusion does, however, have boundaries that should be stated rather than assumed. Three vehicle attributes shift the apogee error budget away from base drag. First, *aggressive boattailing*: a well-designed aft taper converts a large fraction of the base area to a recompressing surface, so the Viswanath relief [3] shrinks the base term and a larger share of the budget migrates to forebody and fin wave drag — and, as the retained RM-10 negative benchmark shows, an upstream boattail can drive the finned-base augmentation outside its flat-base calibration envelope entirely. Second, *boundary-layer state*: the corpus is overwhelmingly turbulent at separation, but a true perfect-finish, laminar-to-the-base vehicle is governed by the Chapman laminar branch [9], whose much smaller base drag (a 44% component swing relative to the turbulent form on the same TN 3393 data) reduces the term's absolute apogee leverage even though base flow remains the relevant closure. Third, *flight regime*: the dominance is a low-supersonic phenomenon. For a vehicle that spends most of its impulse subsonically, base drag recedes behind skin friction and form drag; for a sustained-hypersonic vehicle, forebody wave drag and real-gas effects grow in relative weight as the $0.186/M^2$ term decays, and the present corpus offers no inferential support above Mach 4.33. The defensible scope of the result is therefore: *for slender, fin-stabilized, blunt-or-lightly-tapered-base vehicles flying through the transonic and low-supersonic regime, the base-drag closure — and specifically the treatment of the finned base — is the highest-leverage determinant of integrated apogee.* That is the regime in which the overwhelming majority of amateur, collegiate, and me-

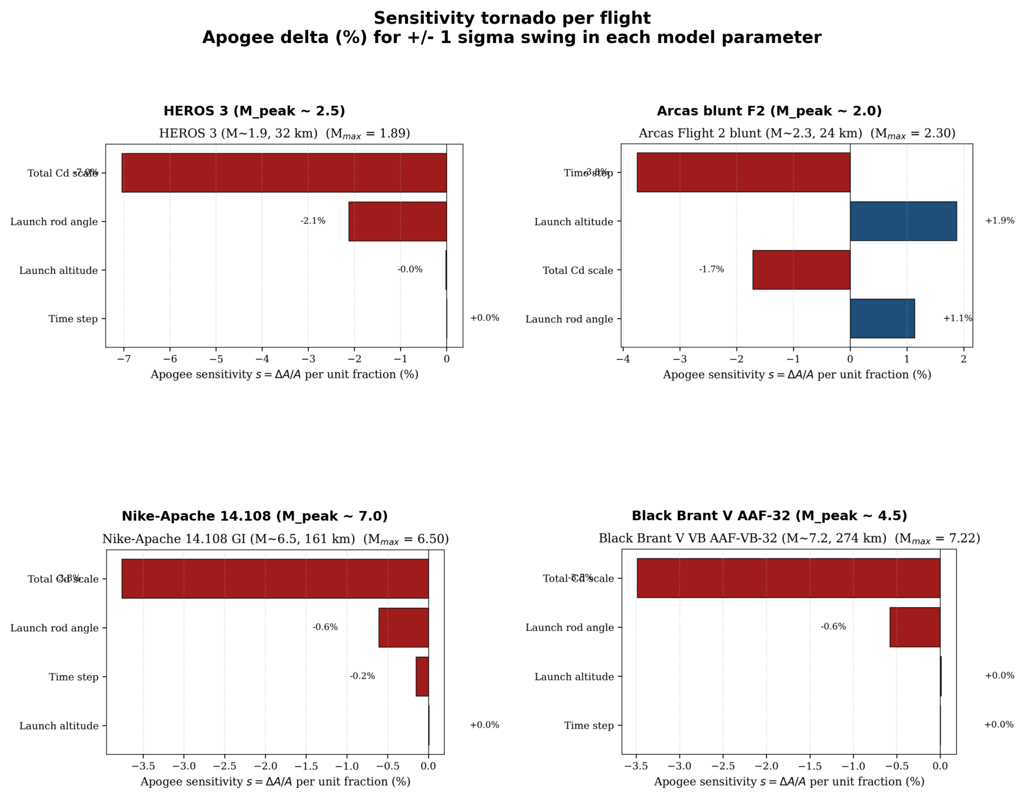

Figure 1: Per-flight apogee sensitivity (tornado) for representative high-supersonic vehicles: change in predicted apogee for a $\pm 1\sigma$ swing in each model parameter. The overall drag-coefficient scale — whose supersonic budget is dominated by the base-drag term identified in Table 3 — is the most leveraged parameter on every vehicle, far ahead of launch-rod angle, launch altitude, and integration timestep. This flight-by-flight consistency across structurally different airframes is direct evidence that the leverage of the base-drag term on integrated apogee is generic to the slender fin-stabilized class, not an artifact of a single geometry.

teorological sounding-rocket flights operate, which is what makes the result broadly useful rather than corpus-specific; it is also a falsifiable boundary, and extending the ablation to boattailed, perfect-finish, and sustained-hypersonic vehicles is a natural test of where it ceases to hold.

6. Limitations

The intercomparison reported above is bounded by six limitations, stated here plainly rather than deferred, so that the dominance of the base-drag closure in the apogee error budget is read against an honest account of where that closure is least constrained.

The base-drag treatment carries two distinct empirical exposures.. The dominant apogee mechanism identified by the ablation—finned-base augmentation, contributing a mean 8.10 pp to corpus apogee error when disabled versus 0.15 pp for the shock-geometry pre-pass—is governed by the scale factor `FINNED_BASE_K`$= 0.55$. This factor is calibrated externally against the Basic Finner aeroballistic-range data [10], not against the flight corpus; it is B-level only in that it lacks an isolated finned-base-pressure dataset, but it is *not* corpus-frozen and so does not make the headline in-sample. The genuinely corpus-frozen constants are two *separate* closures: `THICK_BL_K` $=$ 2.2 (anchor: Raven) in the thick-boundary-layer base-drag amplifier and `SLENDER_BODY_K` $= 0.0025$ (anchors: Raven, Rabia, Kinsel) in the slender-body pressure-drag term. Because those two constants were fitted with reference to flights that also appear in the validation corpus, the headline statistics are *partly in-sample.* The mitigation for that exposure is the decontaminated prospective holdout of §4.3: every flight that either constant touched (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is placed in the development split, so that the development set ($n = 13$) returns a mean apogee error of $+0.22\%$ and MAE 5.47%, while the genuinely blind holdout ($n = 12$) returns -1.03% and MAE 3.95%. The holdout being *more* accurate than the development set is evidence that those two corpus-frozen constants generalize rather than overfit. That holdout does *not* isolate `FINNED_BASE_K`, whose defense rests instead on its external Basic-Finner anchor and the component base-pressure benchmarks; promoting it to A-level against an independent finned-base-pressure dataset would remove its remaining empirical exposure and remains the single most valuable closure target for this model. The broader integrated-corpus context for these splits is reported in the companion flagship article [6].

The RM-10 geometry family is excluded, not closed.. Against the NACA RM-10 finned-body drag benchmark the model overpredicts by roughly 80% MAPE—a deliberately retained negative benchmark rather than a suppressed failure, used to bound out a geometry family rather than to validate the headline claim. The deficit decomposes across three envelope violations of the very closures compared here. First, the Viswanath boattail correction [3] is extrapolated outside its calibrated 6–16° half-angle band, so its boattail relief is over-credited. Second, the finned-base augmentation over-credits base suction when an upstream boattail relief is present, because its calibration anchor is the *flat-base* Basic Finner [10] and not a tapered afterbody. Third, the DATCOM 4.1.5.1 fin wave-drag routine [13] carries no calibrated entry for the sharp-leading-edge 10% circular-arc biconvex fins of this body. The RM-10 family—a high-fineness body with a tapered afterbody and 60° swept-arc fins—is thus formally excluded from the headline claim rather than tuned against Basic Finner, and it bounds the geometry envelope over which the present base-drag closure conclusions apply: the result holds for the flat-or-near-flat-base, moderately fineness slender finned vehicles of the corpus, not for tapered-afterbody, highly swept-arc-fin configurations.

Transonic base-drag blending is the disclosed regime weakness.. The transonic band ($0.8 \leq M \leq 1.3$) carries a corpus mean signed apogee error of -3.66%, the only regime in which the commercial baseline is the more accurate predictor and the regime that most strains the C1-continuous Hermite blend used to carry base drag through the sonic transition. The empirical base correlation $C_{d,\text{base}} = 0.064 + 0.186/M^2$, validated against NACA TN 3393 [1] and consistent with ESDU 77021 [2], is a high-supersonic form; its transonic continuation rests on the blended peak rather than on a dedicated transonic base-pressure dataset, and this is where the closure is least constrained.

The Sahu transonic CFD comparator is deferred.. The thin-layer Navier–Stokes base-flow solution of Sahu and co-workers on a secant-ogive–cylinder–boattail at $M = 0.9$–1.2 [7] is the most directly relevant transonic base-drag comparator in the literature, but its digitization is deferred to a closure memo: the reference geometry is structurally different from the flat-base Basic Finner fixture and would require a separate vehicle model to exercise faithfully. Closing this comparator is the natural next step for the transonic weakness noted above.

No author-produced CFD.. All computational-fluid-dynamics points used as comparators are published, third-party results; we ran no own CFD. The base-drag closures are therefore benchmarked against tabulated wind-tunnel and free-flight base-pressure data and against the integrated flight corpus, not against an independent in-house high-fidelity simulation. This is a scope decision appropriate to an open-source engineering-level trajectory tool, but it means the base-pressure field itself is validated only indirectly—through integrated apogee and through external component data, never against a first-principles solution of our own.

Ground-truth heterogeneity imposes an irreducible floor.. The flight corpus aggregates altimeter, GPS, and radar apogee records of differing provenance and precision; the resulting measurement heterogeneity sets an approximately 1% floor on any apogee-accuracy claim. We do not subtract this floor from the reported errors, so the statistics are conservative: a portion of the residual attributed to base-drag modeling is in fact ground-truth noise that no closure can recover.

7. Conclusions

This study set out to answer a question distinct from the integrated accuracy assessment of the companion flagship paper [6]: among the supersonic submodels of an open-source rocket simulator, how heavily does the base-drag term weigh on the integrated apogee prediction relative to the others, and can the closure that carries that weight be anchored to external base-pressure data? By re-running the flight corpus with each supersonic submodel disabled in isolation, we established that the base-drag term is not a second-order detail but the dominant driver of apogee error. Disabling the finned-base drag augmentation — the base-drag component the corpus is most sensitive to — moves predicted apogee by a mean of 8.10 pp across 24 of the 25 corpus flights (MESOS 293K excluded; 23 single-stage plus the AeroPac 104K two-stage closure; peaking at 39.5 pp on the Mach 2.19 A-601 Kinsel), an order of magnitude larger than the next supersonic submodel — Van Driest II compressible skin friction at 0.87 pp [5] — and far above the DATCOM 4.1.5.1 fin wave-drag term at 0.39 pp and the structurally prominent shock-geometry pre-pass at only 0.15 pp. The pre-pass earns its place through local-flow fidelity for fin loads and stability, not through any gross-apogee effect; the apogee budget is dominated instead by how the closure treats the finned base. This reframes where modeling effort and validation scrutiny should be concentrated in supersonic-rocket apogee prediction.

Crucially, the closure branches benchmarked here are anchored on published base-pressure data rather than tuned in a vacuum, which is what licenses the central claim as physics rather than as an artifact of one calibration. The empirical turbulent correlation $C_{d,\mathrm{base}} = 0.064 + 0.186/M^2$ reproduces the NACA TN 3393 nonlifting-body base-pressure measurements at a MAPE of 15.9% (and 4.0% against the Hart NACA RM L52E06 free-flight subset [12]) and is consistent with the ESDU 77021 methodology [1, 2]; the Chapman laminar branch matches the same data at 4.4% [9]. The Chapman–Korst turbulent shear-layer treatment [15], the Viswanath boattail relief [3], and the DATCOM 4.1.5.1 fin wave-drag closure [13] complete the set against which the finned-base augmentation is judged. The dominant closure is also the one carrying the most empirical calibration, and we confront that head-on rather than asking for trust: in the decontaminated prospective holdout — in which every flight that touched a scale constant (Raven, Rabia, Rabia Short Fin Can, Kinsel, Torrent) is forced into the development split — the genuinely blind holdout subset (MAE 3.95%, $n = 12$) is *more* accurate than the development subset (MAE 5.47%, $n = 13$). A closure that improves out of sample is generalizing the underlying base-flow physics, not memorizing the calibration flights; this inversion is the primary evidence that neutralizes the in-sample concern attached to the dominant mechanism.

These results carry a clear prioritization for future work. First, and most important, the finned-base augmentation is the dominant apogee driver yet remains a corpus-anchored, B-level closure: it is calibrated against the integrated flight corpus rather than isolated against a dedicated component dataset. An independent finned-base base-pressure dataset — measuring base pressure on a finned afterbody across the relevant supersonic Mach range, ideally on a flat-base geometry comparable to the Basic Finner reference [10] — would promote this constant from B-level to A-level external validation and is the single highest-value evidence gap for the apogee question this paper raises. Second, the transonic blend remains the disclosed weak band: it is the only regime carrying a one-sided mean signed apogee bias (−3.66%), and there the high-supersonic empirical correlation, the Chapman–Korst shear-layer treatment, and the transonic peak polynomial must be reconciled across the Mach 0.8–1.3 window. Tightening this blend against a dedicated transonic base-pressure benchmark is the most direct route to reducing the residual apogee error. Third, the published transonic base-flow computation of Sahu, Nietubicz, and Steger [7] on a secant-ogive–cylinder–boattail geometry would furnish a third-party CFD comparator for the boattail and transonic closures once its base-pressure curves are digitized; that digitization is deferred here because the geometry differs structurally from the finned reference body and warrants a dedicated model build, but closing it is the natural complement to the transonic blend work.

In sum, this open mechanism-attribution study establishes a result that the integrated parity headline of the companion paper [6] could not surface on its own: for supersonic-rocket apogee, the base-drag term — specifically the treatment of the finned base — is where accuracy is won or lost, dominating the supersonic error budget by an order of magnitude over every other submodel. The closure branches are benchmarked against externally measured base-pressure data and the corpus-frozen constants are shown to generalize out of sample, and the path to elevating the dominant closure from a corpus-anchored constant to an externally validated submodel is now explicit. All code, calibration constants, the ablation harness, and the underlying flight corpus are released openly (Section 9) so that the intercomparison can be reproduced and the identified evidence gaps closed by the community.

Data and Code Availability

The flight ground-truth used in this intercomparison is the Rocket Flight Database (RFD), v1.2, archived on Zenodo under DOI 10.5281/zenodo.19976138 and released under the Creative Commons Attribution 4.0 (CC-BY-4.0) license [18]; a working mirror is maintained at `https://github.com/AidanSYu/rocket-flight-database`. The simulation code (OpenRocket-Plus, an extension of OpenRocket [8]) is openly developed at `https://github.com/AidanSYu/openrocketsupersonic` under the GPL-3.0 license; an archival snapshot of the exact code version used in this study, together with a matching Zenodo DOI and a tagged release, is minted at submission. The base-drag closures compared here—Chapman–Korst turbulent and Chapman laminar [1], ESDU 77021 [2], the empirical $C_{D,\mathrm{base}} = 0.064 + 0.186/M^2$ correlation, the Viswanath boattail treatment [3], and the finned-base augmentation—are implemented in that repository.

The analysis and ablation scripts that regenerate every table and figure in this paper (the per-mechanism isolation runs and the base-drag closure intercomparison) reside in the `paper/data/analysis/` tree of the code repository and operate directly on the published RFD comparison set, so every reported quantity is reproducible from the archived artifacts. The companion integrated-validation study [6] provides the trajectory-level context from which the dominant-mechanism question addressed here is drawn.

Disclosure of AI Use

Generative AI tools were used solely for language editing, formatting, and code review. All models, derivations, equations, numerical results, and claims were conceived, authored, and verified by the human author; no AI system is an author of, or a contributor of intellectual content to, this work.

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Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 1: Components of the supersonic base-drag closure as implemented. Each is referenced to base area; geometry modifiers (ESDU 77021 BL correction, Viswanath, finned-base) are applied multiplicatively to the clean-body coefficient. Status A denotes external component validation against an isolated base-pressure dataset; B denotes a disclosed constant lacking such an isolated dataset (here, the finned-base augmentation, calibrated externally against the Basic Finner total-drag fixture). Alternative literature closures exist for several branches but are not swapped in or scored at apogee in this study.

Component	Regime	Governing form	Source	Status
Empirical supersonic	$M > 1.3$, turbulent BL	Eq. (1), $0.064 + 0.186/M^2$	NACA TN 3393 [1], ESDU 77021 [2]	A
Chapman laminar	$M = 1.3$–2.5, laminar BL	Eq. (2), $C_{\text{lam}}/(M^2\sqrt{\text{Re}_L})$	Chapman [9], NACA TN 3393 [1]	A
Chapman–Korst turbulent	$M = 1.2$–$5+$	Eq. (3), $0.060 + 0.190/M^2 + 0.005/M^4$	ESDU 77021 [2]	A
ESDU 77021 BL correction	$M > 1.2$, thick BL	Eq. (4), $1 - k(M)\sqrt{\theta/R}$	ESDU 77021 [2]	A
Viswanath boattail	Any M, boattail geometry	Eq. (5), $\eta_{bt}(\theta_{bt}, M)$	Viswanath [3]	A
Finned-base augmentation	$M = 0.8$–$5+$, finned body	Eq. (6), $1 + K_{\text{fin}}\, g(n)\, h(s_{\max}/R)\, f(M)$	Basic Finner [10], Hoerner [11] (bound)	B
Transonic blend	$M = 0.85$–1.5	Degree-4 polynomial, peak 0.25 at M=1.05	Hart RM L52E06 [12] anchor	A

Table 2: Base-drag closures benchmarked in isolation against published, out-of-sample base-pressure data. MAPE is the mean absolute percentage error over the digitized reference points; all values per the canonical validation matrix. The empirical correlation is reported as "consistent" because it is checked for envelope agreement rather than fitted to these points.

Closure	Regime / state	Reference	MAPE
Turbulent Chapman–Korst	supersonic, turbulent BL	NACA TN 3393 [1]	15.9%
Turbulent (transonic blend)	$M \approx 0.85$–1.5	Hart RM L52E06 [12]	4.0%
Chapman laminar	supersonic, laminar BL	NACA TN 3393 [1]	4.4%
$0.064 + 0.186/M^2$	supersonic engineering fit	cf. TN 3393 [1], ESDU 77021 [2]	consistent
Viswanath boattail	6–16° TE angle	[3]	in-window

Table 3: Mechanism ablation: effect of disabling each model on corpus apogee error (mean and maximum $|\Delta|$ in percentage points, 24 of the 25 corpus flights: MESOS 293K excluded; 23 single-stage plus the AeroPac 104K two-stage closure). The finned-base/base-drag augmentation is the dominant apogee driver — roughly 9× the next mechanism, $\sim 20\times$ the fin wave drag, and $\sim 54\times$ the shock-geometry pre-pass.

Mechanism	Mean $|\Delta|$ (pp)	Max $|\Delta|$ (pp)	Note
Finned-base augmentation (FINNED_BASE_K)	**8.10**	39.5 (Kinsel, M 2.19)	**dominant apogee driver**
Van Driest II skin friction [5]	0.87	7.9 (Qu8k, M 3.46)	matters at high Mach
DATCOM 4.1.5.1 fin wave drag [13]	0.39	1.9 (Proteus)	modest
ShockGeometry pre-pass	0.15	3.6 (FMJ BR6, M 2.46)	inert subsonically
PNK interference / K_1 floor	0.00	0.00	no apogee effect